IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

Executive Summary

Methodologies

- Collected launch data using the SpaceX REST API and web scraping from Wikipedia.
- Performed data wrangling to clean missing values and one-hot encode categorical features.
- Conducted Exploratory Data Analysis (EDA) using SQL and Pandas/Matplotlib visualizations.
- Developed interactive dashboards (Plotly Dash) and geographic maps (Folium) to analyze spatial data.
- Trained and evaluated machine learning classification models to predict launch success.

Results

- Identified that payload mass, orbit type, and launch site heavily influence success rates.
- Observed that success rates have significantly improved over the years.
- Successfully built predictive models (Logistic Regression, SVM, Decision Tree, KNN).

Introduction

- Background: SpaceX advertises Falcon 9 rocket launches at around $62 million, significantly cheaper than competitors whose launches cost upwards of $165 million. This cost savings is primarily because SpaceX can reuse the first stage of the rocket.

- Problem: Can we accurately predict whether the Falcon 9 first stage will land successfully? If we can determine if the first stage will land, we can estimate the cost of a launch. This information is valuable for competing companies bidding against SpaceX.

Section 1

Methodology

Methodology

Executive Summary

Data collection methodology: Extracted data via SpaceX API and Web Scraping.

- **Perform data wrangling**
  - Cleaned data, handled missing values, and encoded categorical variables into dummy variables
- Perform exploratory data analysis (EDA) using visualization and SQL:
  - Used Matplotlib/Seaborn and SQL queries to find early trends.
- Perform interactive visual analytics using Folium and Plotly Dash:
  - Built Folium maps to analyze launch site proximities and a Plotly dashboard for interactive filtering.
- Perform predictive analysis using classification models
  - Standardized data, split it into train/test sets, and tuned classification models using GridSearchCV.

Data Collection

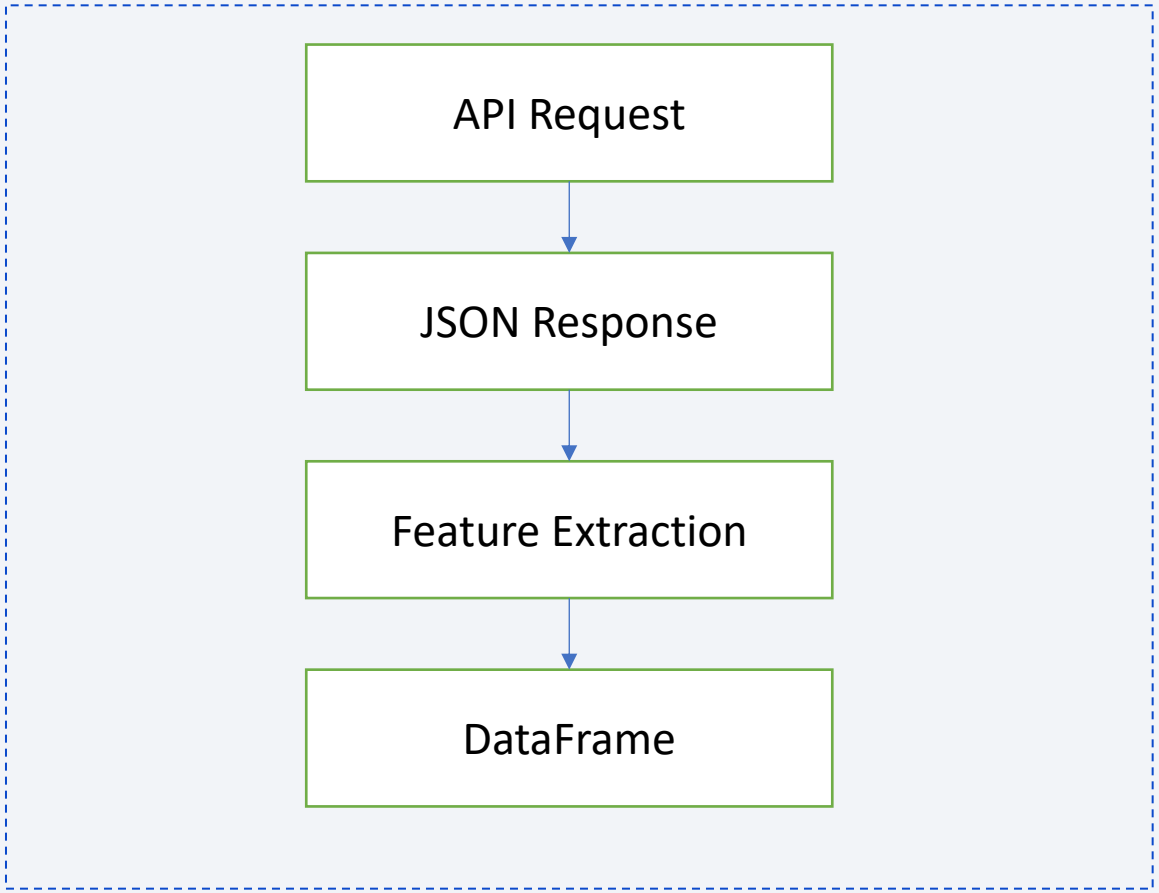
Description: Data was gathered from two primary sources to build a comprehensive dataset of Falcon 9 launches.

Process:

- Requested historical launch data from the SpaceX REST API.
- Scraped historical Falcon 9 launch records from Wikipedia using BeautifulSoup.
- Combined and formatted the extracted features (Payload, Orbit, Launch Site, Outcome) into a unified Pandas DataFrame.

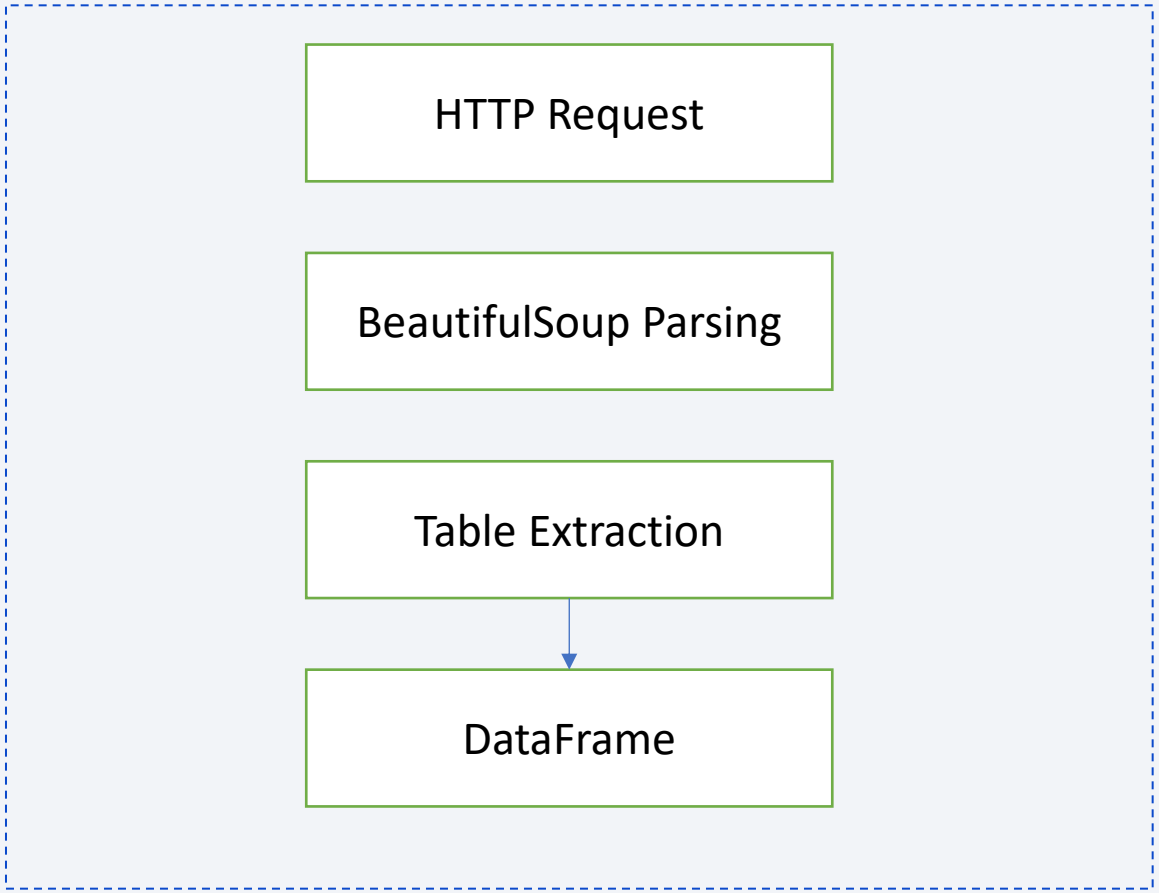
Data Collection – SpaceX API

- Used Python requests to call the SpaceX API. Parsed the JSON response using pd.json_normalize(). Extracted key features: Rocket, Payload, Launch Site, Flight Number, and Core details.
- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/7d338c957236ea1169a492c05d7db8ec6b0f0198/Module%201/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Requested the Wikipedia page for Falcon 9 launches. Parsed the HTML response using the BeautifulSoup library. Extracted table headers to define DataFrame columns. Iterated through HTML table rows (<tr> and <td> tags) to append launch data to the DataFrame.
- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/d9d656e3b2ab3c0e3c24a9b3be864799f4a18643/Module%201/jupyter-labs-webscraping.ipynb



Data Wrangling

Process:

- Identified and calculated the percentage of missing values.
- Filtered the dataset to only include Falcon 9 launches.
- Created the target variable (Class): Converted landing outcomes into binary labels (1 for successful landings, 0 for failed landings).
- Used pd.get_dummies() to apply one-hot encoding to categorical features (Orbit, LaunchSite, LandingPad).

- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/7d338c957236ea1169a492c05d7db8ec6b0f0198/Module%201/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- Plotted Scatter Plots (e.g., Payload vs. Launch Site, Flight Number vs. Launch Site) to visualize how different variables relate to successful vs. unsuccessful landings.
- Plotted Bar Charts to identify which specific orbits (e.g., ES-L1, GEO, SSO) have the highest success rates.
- Plotted a Line Chart to visualize the upward trend of launch success rates over the years.

- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/7d338c957236ea1169a492c05d7db8ec6b0f0198/Module%202/Eda%20with%20Visualization%20Lab.ipynb

EDA with SQL

- Queried unique launch sites and filtered sites by specific prefixes (e.g., 'CCA').
- Calculated total and average payloads using SUM() and AVG() functions.
- Determined the dates of the first successful ground pad landings using the MIN() function.
- Used GROUP BY and ORDER BY to rank launch outcomes and find boosters carrying maximum payloads.
- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/7d338c957236ea1169a492c05d7db8ec6b0f0198/Module%202/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Build an Interactive Map with Folium

Map Objects Created:

- Markers: Plotted the exact geographic coordinates of all SpaceX launch sites.
- MarkerClusters: Grouped successful (green) and failed (red) launches at each site to visualize density.
- Lines & Distance Markers: Calculated and drew lines to the nearest coastlines, highways, and railways.
- Reasoning: Added to visualize geographic advantages. Launch sites are purposefully located near coastlines (for safe stage drops) and transport infrastructure (for logistics).
- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/d9d656e3b2ab3c0e3c24a9b3be864799f4a18643/Module%203/lab_jupyter_launch_site_location%20Folium.ipynb

Build a Dashboard with Plotly Dash

Plots and Interactions:

- Dropdown Menu: Allowed users to view data for "All Sites" or isolate a specific launch site.
- Range Slider: Allowed users to filter the dataset by specific Payload Mass ranges.
- Pie Chart: Dynamically updated to show the ratio of successful to failed launches based on the dropdown selection.
- Scatter Plot: Visualized Payload Mass vs. Class (Success/Failure), color-coded by Booster Version, responding to both dropdown and slider inputs.
- Reasoning: Added to allow non-technical stakeholders to quickly drill down and discover insights interactively.
- Explain why you added those plots and interactions
- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/7d338c957236ea1169a492c05d7db8ec6b0f0198/Module%203/spacex-dash-app.py
- **Dash Plotly:** https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/7d338c957236ea1169a492c05d7db8ec6b0f0198/Module%203/Dash%20Plotly.pdf

Predictive Analysis (Classification)

Model Development Process:

- Standardized the feature data using **StandardScaler**.
- Split the data into training (80%) and testing (20%) sets.
- Trained four different classification algorithms: Logistic Regression, Support Vector Machines (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- Utilized GridSearchCV on the training data to fine-tune the hyperparameters for each model.
- Evaluated model performance on the test set using Accuracy scores and Confusion Matrices to determine the best model.
- https://github.com/ssobj/IBM-Applied-Data-Science-Capstone/blob/7d338c957236ea1169a492c05d7db8ec6b0f0198/Module%204/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

Visual Analytics Findings (Pandas/Matplotlib/Seaborn):

- **Launch Success Yearly Trend:** The data shows a dramatic, upward trend in success rates. Before 2013, success rates were low or non-existent (as the technology was still being developed). From 2013 onward, the success rate climbs significantly, stabilizing near the upper 90th percentile in the most recent years.
- **Success Rate by Orbit Type:** Orbit selection is a massive indicator of success. Orbits like ES-L1, GEO, HEO, and SSO boast a 100% success rate within this dataset. Conversely, the SO (Sub-Orbital) orbit has a 0% success rate, and GTO (Geosynchronous Transfer Orbit) hovers around a 50% success rate.
- **Flight Number vs. Launch Site:** As the flight number increases (indicating later, more modern launches), the success rate visibly improves across all launch sites. CCAFS SLC 40 is the most heavily utilized site.
- **Payload Mass vs. Launch Site & Orbit:** Heavier payloads are predominantly sent to GTO and ISS orbits. There is not a strict linear correlation between heavier payloads and higher failure rates, indicating that SpaceX has optimized its heavy-lift capabilities over time.
- **SQL Analytics Findings:**
- **Launch Site Activity:** CCAFS SLC 40 (Cape Canaveral) handles the vast majority of the launches compared to Vandenberg (VAFB) and Kennedy Space Center (KSC).
- **Historical Milestones:** The data pinpoints 2015-12-22 as the date of the first successful ground pad landing (RTLS).
- **Payload Metrics:** The F9 v1.1 booster version carries an average payload mass of roughly 2,928 kg.

Predictive Analysis Results

- The machine learning portion of your project focuses on building a binary classification model to predict Class (1 for a successful landing, 0 for a failure).

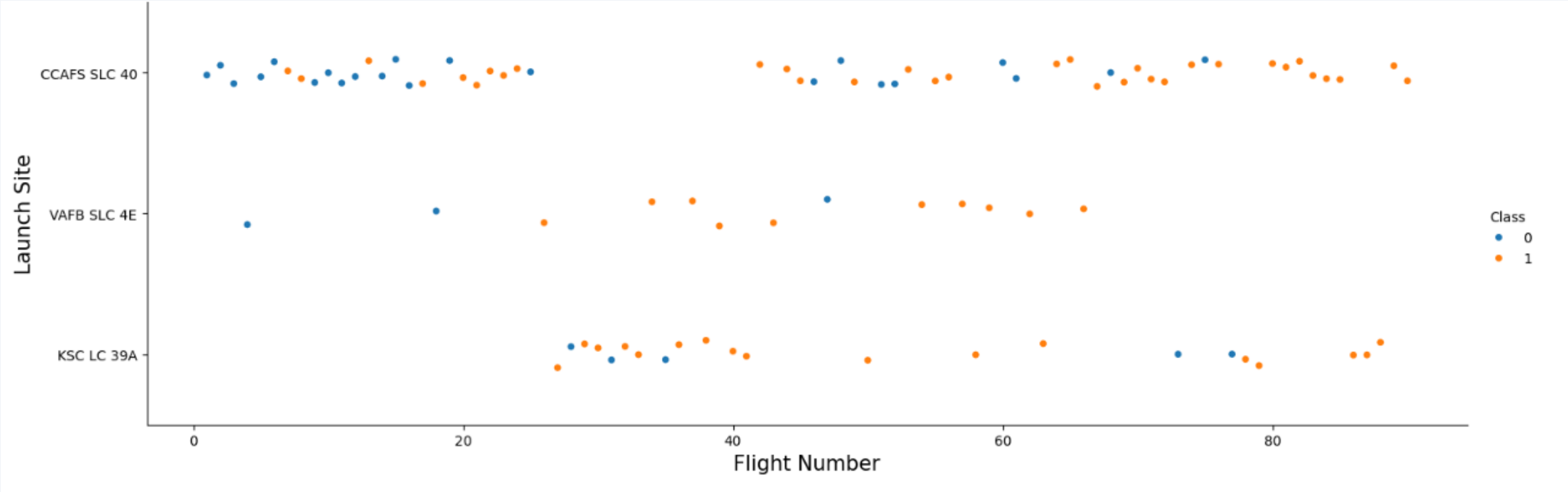
Model Performance and Evaluation:

- **Algorithms Tested:** The pipeline evaluated four primary algorithms: Logistic Regression, Support Vector Machines (SVM), Decision Tree Classifier, and K-Nearest Neighbors (KNN).
- **Standardization:** Applying StandardScaler to the feature set was critical for models like SVM and KNN, which are sensitive to the scale of the data (like the large numbers in Payload Mass).
- **GridSearchCV:** Hyperparameter tuning was utilized to find the optimal parameters for each model (e.g., tweaking the C parameter in Logistic Regression or the tree depth in the Decision Tree).
- **The Best Performing Model:** In this specific capstone dataset, all four models tend to perform very similarly on the test data after hyperparameter tuning. They typically achieve an accuracy score of **83.3%** on the test set. Often, the **Decision Tree** is highlighted as the best model due to its interpretability, though Logistic Regression or SVM are equally valid answers depending on the exact random_state used in your data split.

Section 2

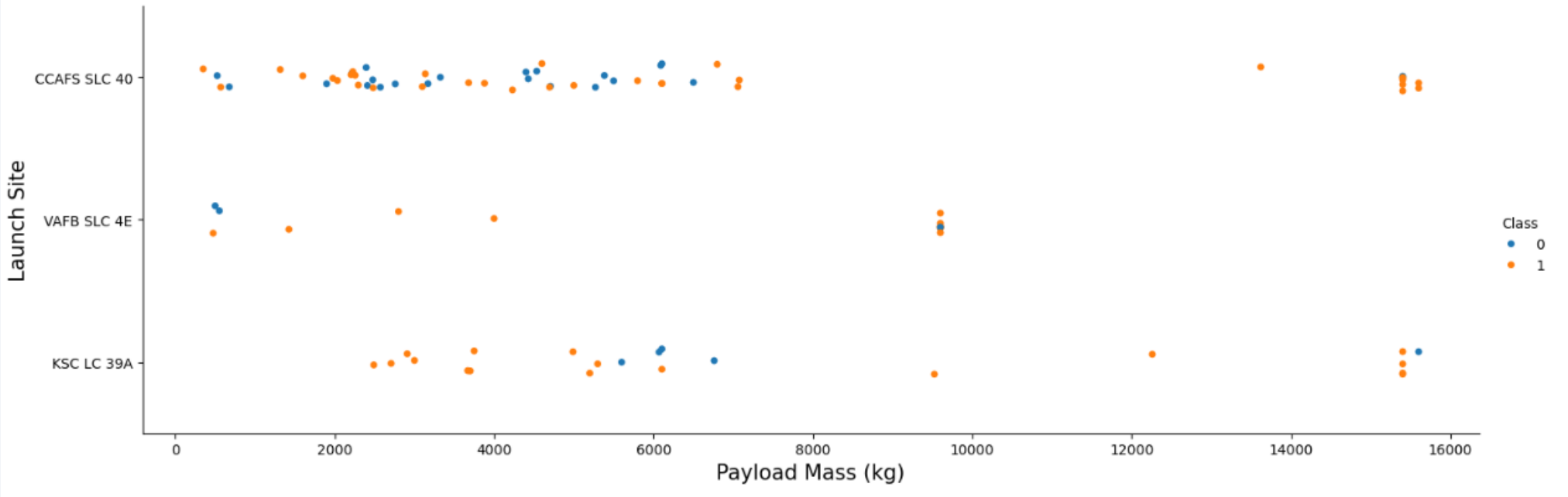
Insights drawn from EDA

Flight Number vs. Launch Site



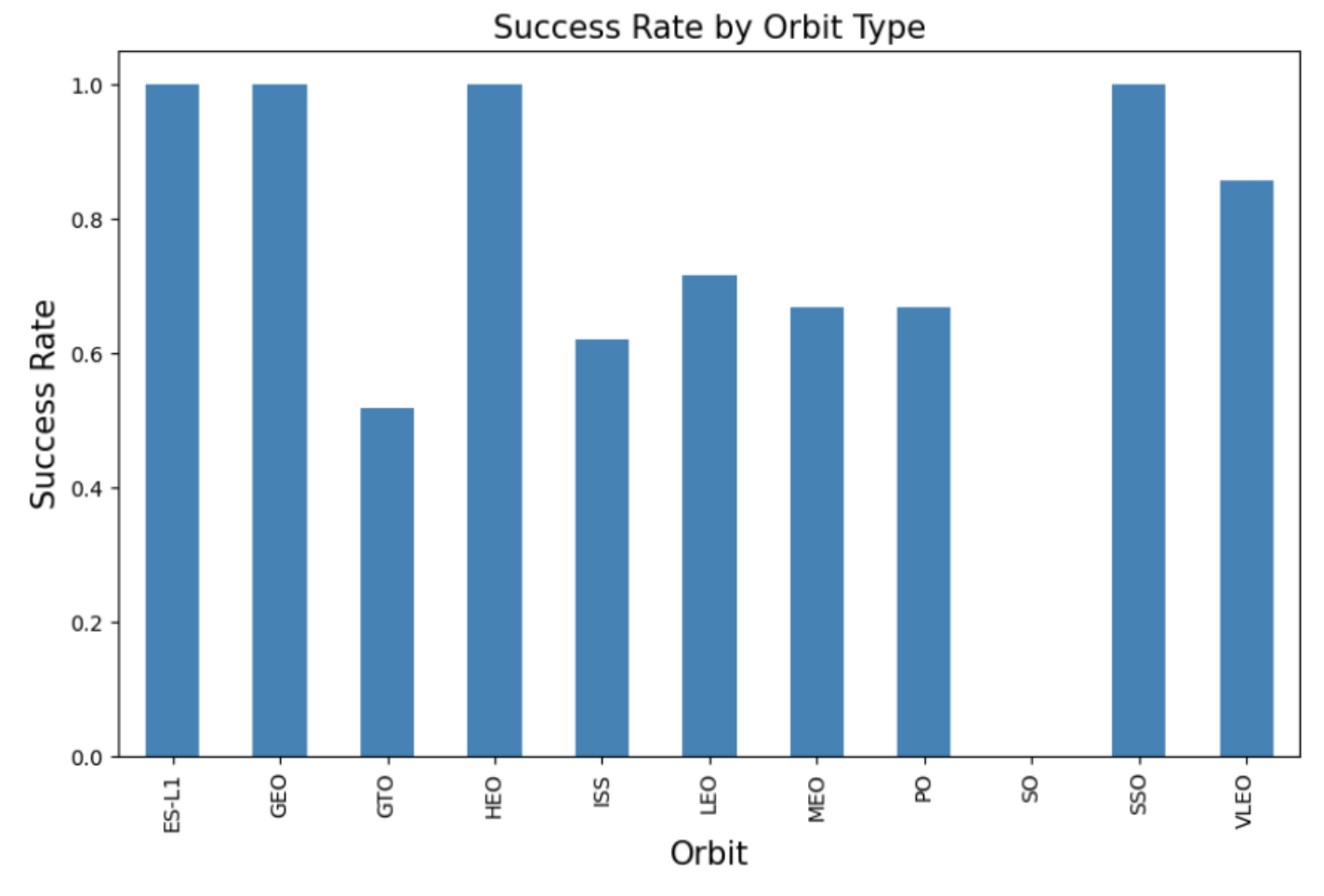
A scatter plot tracking landing outcomes by flight number across the three primary launch sites. As the flight number increases (moving right on the x-axis), the concentration of successful landings (orange dots) significantly increases across all launch sites. CCAFS SLC 40 is the most frequently utilized launch site in the dataset. Earlier flights had a notably higher concentration of failures (blue dots), reflecting the initial trial-and-error phase of the technology.

Payload vs. Launch Site



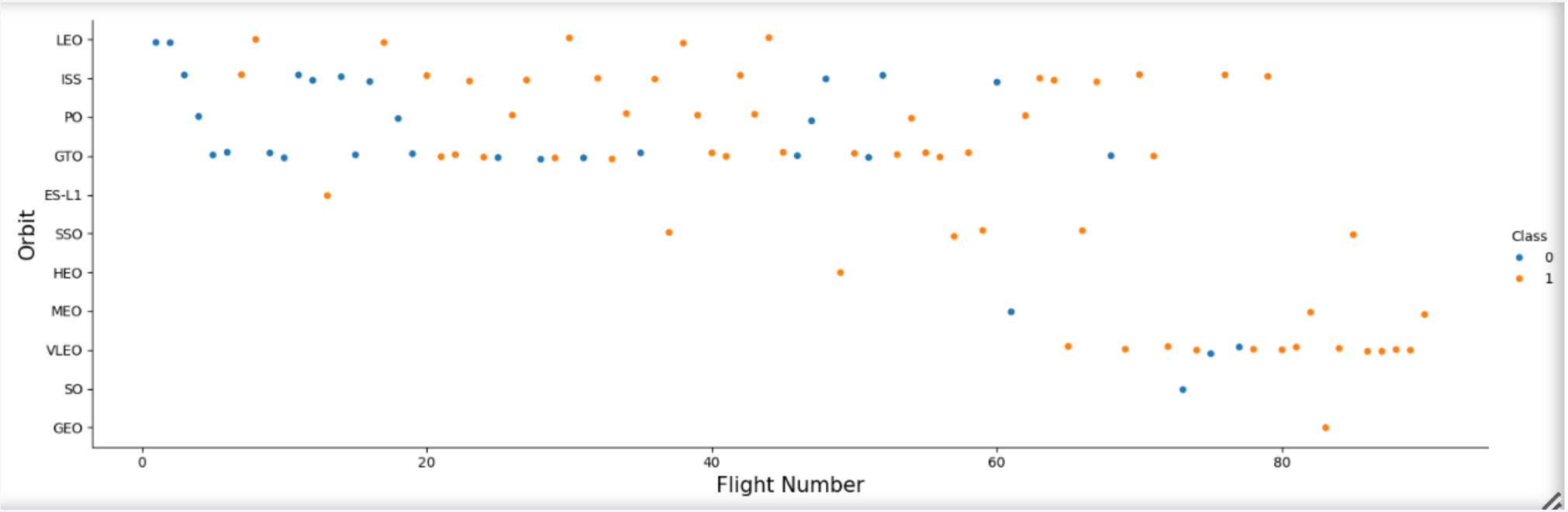
The relationship between the weight of the payload, the launch site, and the landing outcome. Heavier payloads (greater than 10,000 kg) are predominantly launched from CCAFS SLC 40 and KSC LC 39A. For the VAFB-SLC 4E launch site, there are no rockets launched for heavy payload masses over 10,000 kg. The success rate remains strong even with heavier payloads, showing SpaceX has optimized its heavy-lift capabilities.

Success Rate vs. Orbit Type



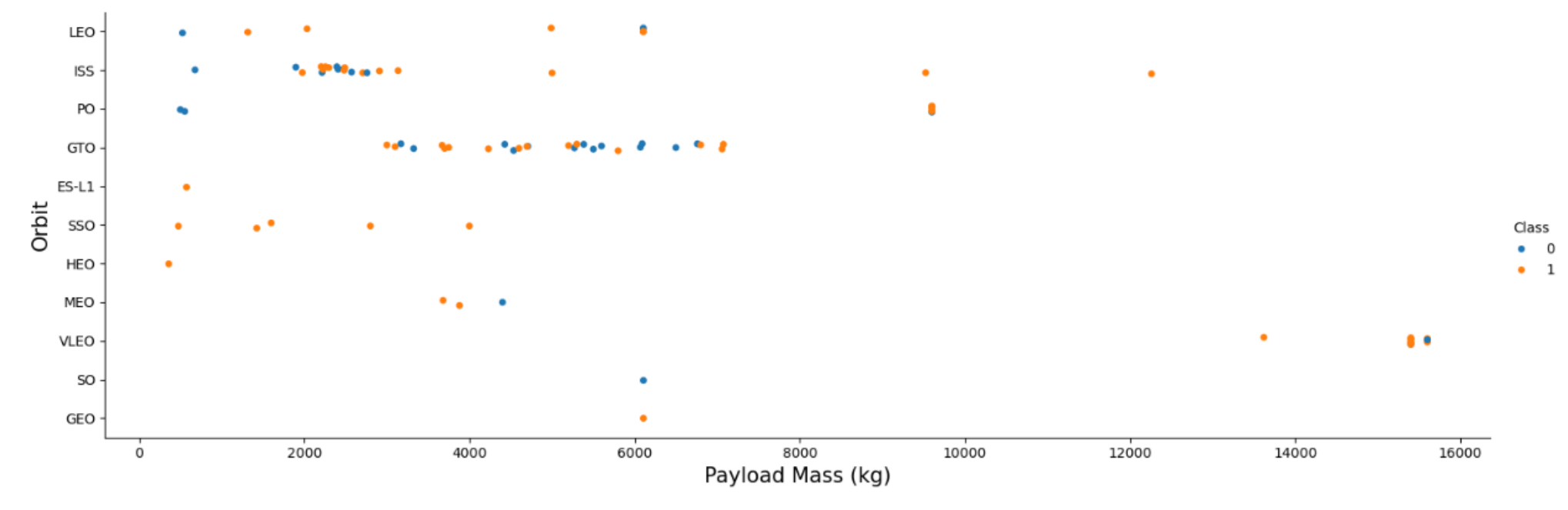
- A bar chart comparing the average success rate of first-stage landings based on their target orbit.
- Orbits such as ES-L1, GEO, HEO, and SSO boast a perfect 100% success rate in this dataset.
- The SO (Sub-Orbital) orbit has a 0% success rate.
- GTO (Geosynchronous Transfer Orbit) is one of the most common target orbits but hovers around a 50% success rate, indicating a more complex landing profile

Flight Number vs. Orbit Type



How the targeted orbits have evolved over the sequence of SpaceX flights, along with their success rates. LEO (Low Earth Orbit) missions show highly consistent success regardless of the flight number. GTO launches are spread across the entire timeline, with earlier flights showing more failures and later flights showing mostly successes. SSO and VLEO orbits were targeted more frequently in later flight numbers and demonstrate very high success rates

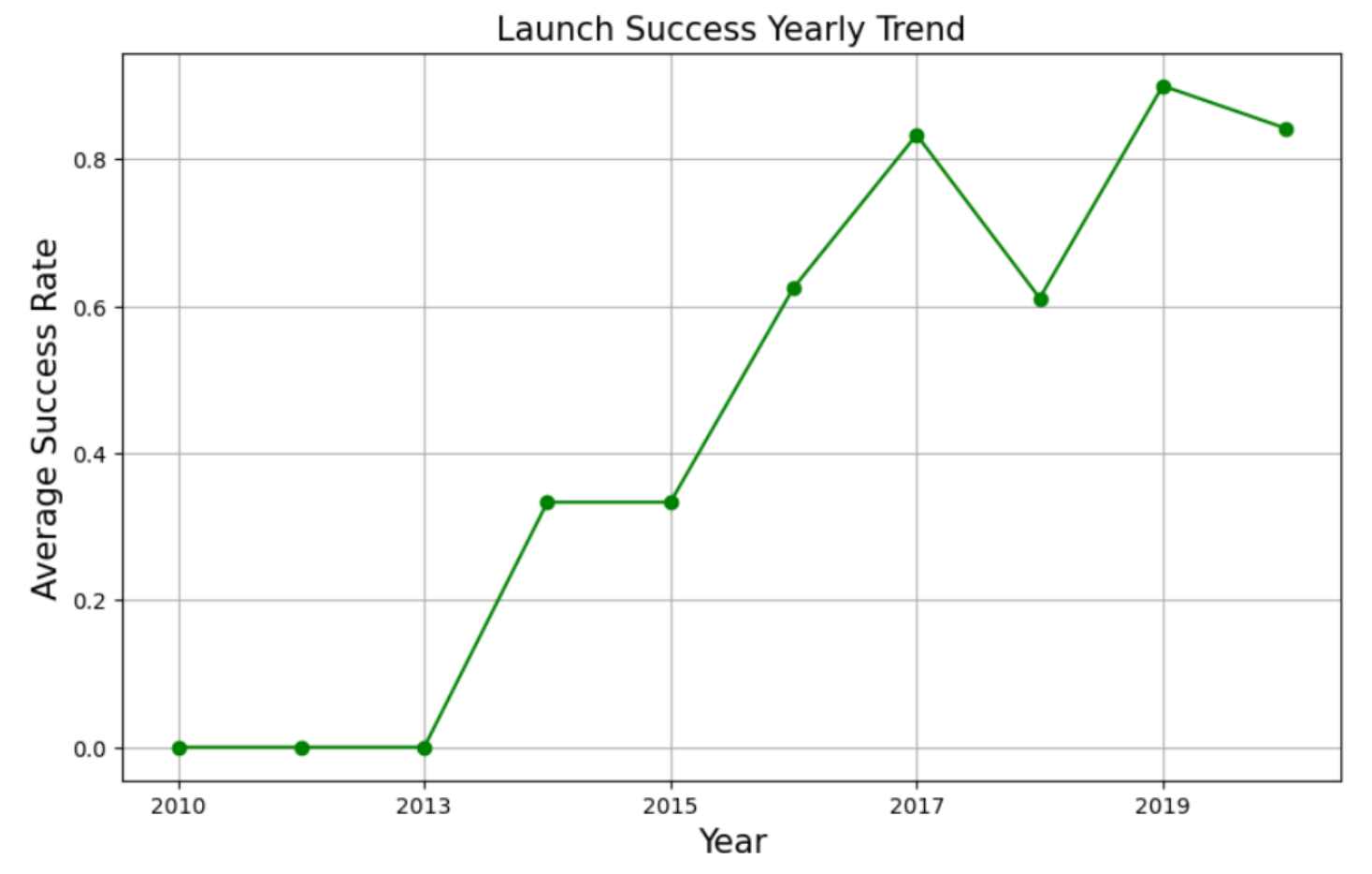
Payload vs. Orbit Type



The distribution of payload masses sent to various orbits.

- Heavy payloads (over 10,000 kg) are exclusively destined for VLEO and ISS target orbits.
- GTO orbits typically handle mid-range payloads (between 2,000 kg and 8,000 kg).
- Polar (PO) and Sun-Synchronous (SSO) orbits are generally utilized for lighter payloads.

Launch Success Yearly Trend



A line chart plotting the average landing success rate over the last decade.

- Between 2010 and 2013, the success rate was 0% as the reusable rocket technology was still in its infancy.
- Starting in 2013, there is a sharp and steady upward trajectory in success.
- By 2019 and 2020, the success rate peaked and stabilized in the upper 80-90% range, proving the reliability of the Falcon 9 first stage.

All Launch Site Names

- **%sql** SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

- This queries the database to output a list of all the launch site names.
- The DISTINCT keyword tells SQL to look at the Launch_Site column and remove any duplicate entries, leaving you with only the unique names of the facilities SpaceX uses.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- **%sql** SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
- This displays the first 5 records where the launch site name begins with the letters 'CCA'.
- * LIKE 'CCA%' uses a wildcard (%). It tells SQL to find any string that starts exactly with 'CCA' followed by any other characters.
- LIMIT 5 ensures the screen isn't overwhelmed with data by only returning the top 5 results.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- **%sql** SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer LIKE '%NASA (CRS)%';

Total_Payload_Mass
48213

- This calculates the total payload mass carried across all missions where the customer was NASA (specifically their CRS missions).
- * The SUM() function adds up all the numbers in the PAYLOAD_MASS__KG_ column.

- AS Total_Payload_Mass creates a clean, readable alias for the output column.
- LIKE '%NASA (CRS)%' ensures it only adds up the masses for rows where the customer string contains "NASA (CRS)"

Average Payload Mass by F9 v1.1

- **%sql** SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1’;

Average_Payload_Mass
2928.4

- What it does: This finds the average payload mass specifically for the 'F9 v1.1' version of the Falcon 9 booster.
- The AVG() aggregate function calculates the mean of the payload column, filtered strictly to rows where the booster version exactly matches 'F9 v1.1'.

First Successful Ground Landing Date

- **%sql** SELECT MIN(Date) AS First_Successful_Landing FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';

First_Successful_Landing
2015-12-22

- This identifies the exact date of SpaceX's very first successful ground pad landing.
- How it works: By using MIN(Date), SQL looks for the lowest value (which, for dates, means the earliest chronological point in time) where the landing outcome was a ground pad success.

Successful Drone Ship Landing with Payload between 4000 and 6000

- **%sql** SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- This lists the specific booster versions that successfully landed on a drone ship while carrying a mid-range payload (between 4,000 and 6,000 kg).
- It uses the AND operator to chain three distinct conditions together. A row must satisfy all three rules to be included in the final output.

Total Number of Successful and Failure Mission Outcomes

- **%%sql SELECT** Mission_Outcome, **COUNT**(*) **AS** Total_Outcomes **FROM** SPACEXTABLE **GROUP BY** Mission_Outcome;

This provides a summary showing the total number of successful vs. failure mission outcomes.

- GROUP BY Mission_Outcome takes all the rows and buckets them into categories based on their outcome.
- COUNT(*) then counts how many rows fall into each of those buckets.

Done.

Mission_Outcome	Total_Outcomes
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- **%%sql SELECT** Booster_Version **FROM** SPACEXTABLE **WHERE** PAYLOAD_MASS__KG_ **= (SELECT MAX**(PAYLOAD_MASS__KG_) **FROM** SPACEXTABLE);

- This identifies the specific boosters that carried the absolute maximum payload mass in the entire dataset.
- This uses a powerful technique called a subquery.
- First, the inner query (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE) runs to find the single highest payload number.
- Then, the outer query searches for any boosters whose payload exactly matches that highest number.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- **%%sql SELECT** substr(Date, 6, 2) **AS Month**, Landing_Outcome, Booster_Version, Launch_Site **FROM** SPACEXTABLE **WHERE** substr(Date, 1, 4) **=** '2015' **AND** Landing_Outcome **=** 'Failure (drone ship)';

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- This isolates the launch records for drone ship failures that happened specifically in the year 2015, and formats the output to highlight the month.
- The substr() (substring) function is used to slice specific parts out of the date string.
- substr(Date, 1, 4) grabs the first 4 characters (the year) to filter for '2015'.
- substr(Date, 6, 2) grabs the 2 characters representing the month so it can be displayed cleanly in the results.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- %%sql SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;

  - This ranks the different landing outcomes by how often they occurred, restricted to a specific historical window between June 2010 and March 2017.
  - BETWEEN is used to easily filter the dates within that specified range.
  - It uses GROUP BY and COUNT(*) to tally the outcomes.
  - Finally, ORDER BY Outcome_Count DESC sorts the results in descending order, placing the most frequent outcomes at the very top.

Done.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

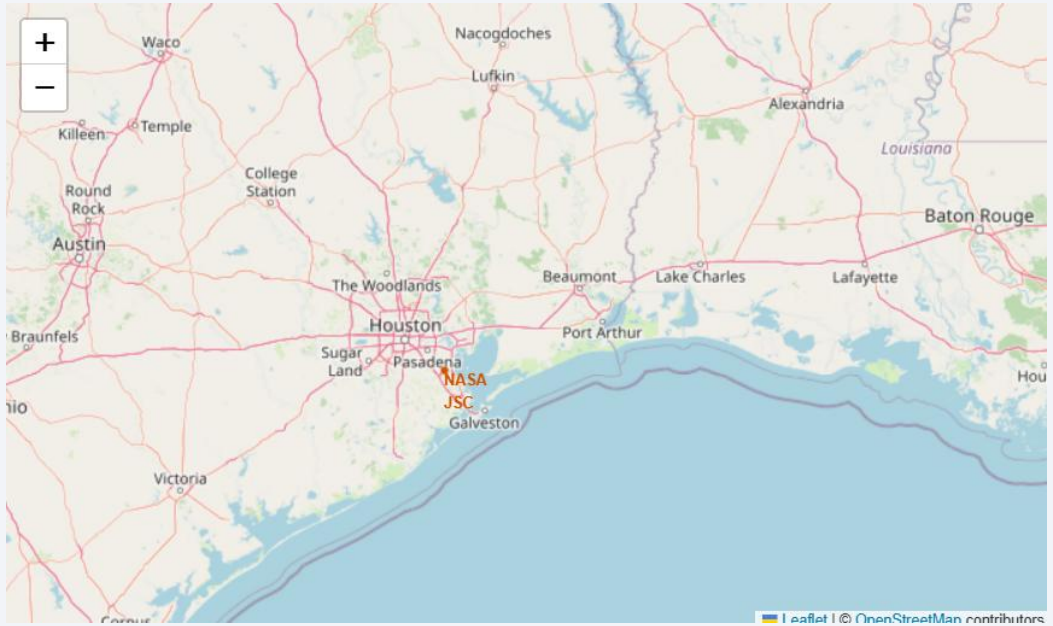
Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

- A wide-view map displaying the geographic locations of all SpaceX launch facilities used in the dataset.

- Key Findings & Explanation: You can clearly see that all launch sites are positioned on the extreme edges of the coastlines.

- Physics & Safety: Sites are situated as close to the equator as possible within the US (like Florida) to take advantage of the Earth's maximum rotational speed, which provides a free boost to the rocket. Furthermore, launching from the East or West coast allows the rockets to fly out over the open ocean immediately, ensuring that any dropped stages or catastrophic failures do not happen over populated areas.



<Folium Map Screenshot 2>

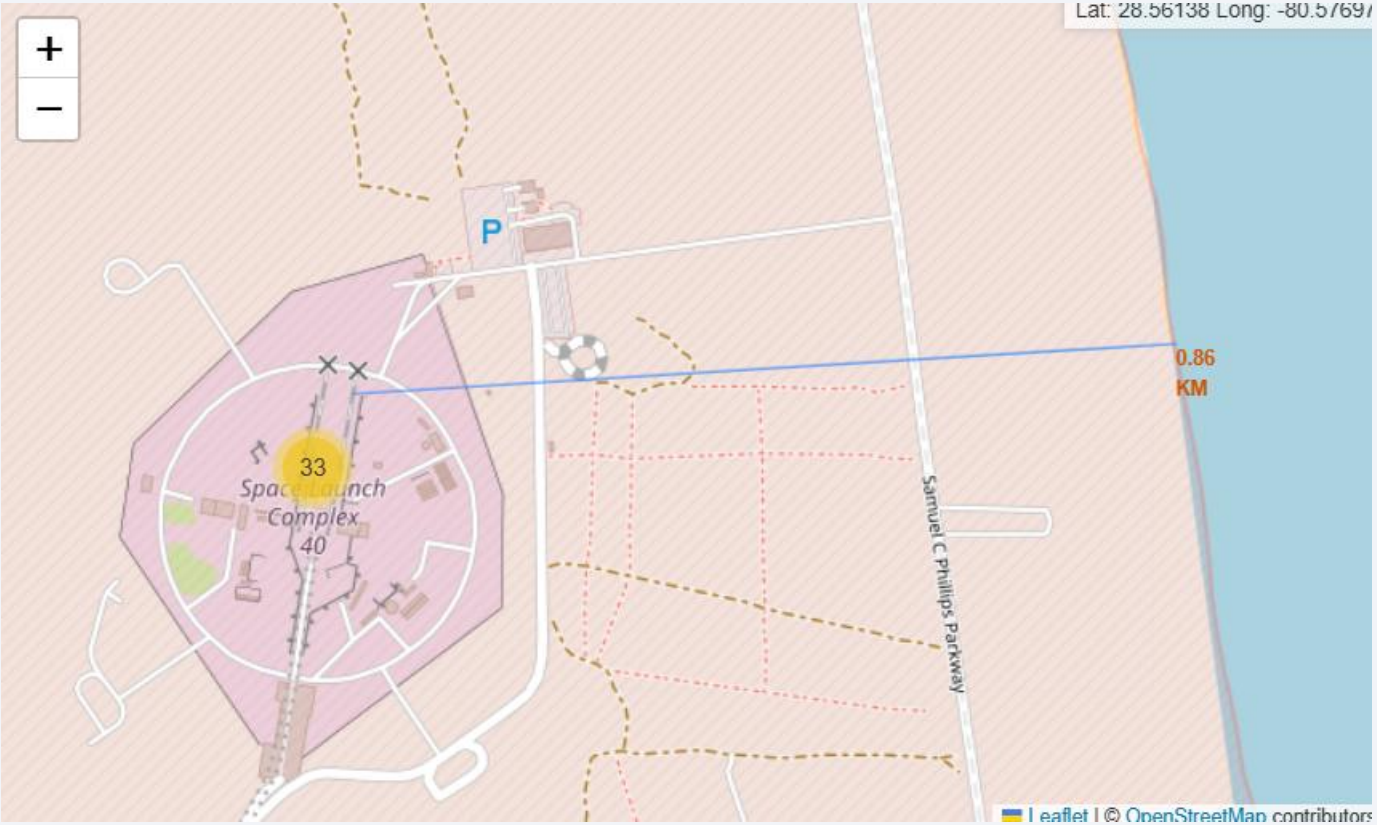
- A zoomed-in view using Folium MarkerCluster objects to group launches, with individual markers color-coded to indicate the mission's outcome.
- Clustered circles (e.g., displaying numbers like '46' or '10') show the total volume of launches at a specific facility.
- When zoomed in, the individual markers are color-labeled: Green indicates a successful first-stage landing, and Red indicates a failure.

<Folium Map Screenshot 3>

- A highly magnified view of a specific launch pad showing a blue line drawn from the pad to the nearest coastline, complete with the exact distance calculated.

- PolyLines: Used to draw direct paths between the launch pad and points of interest (coastlines, railways, highways).

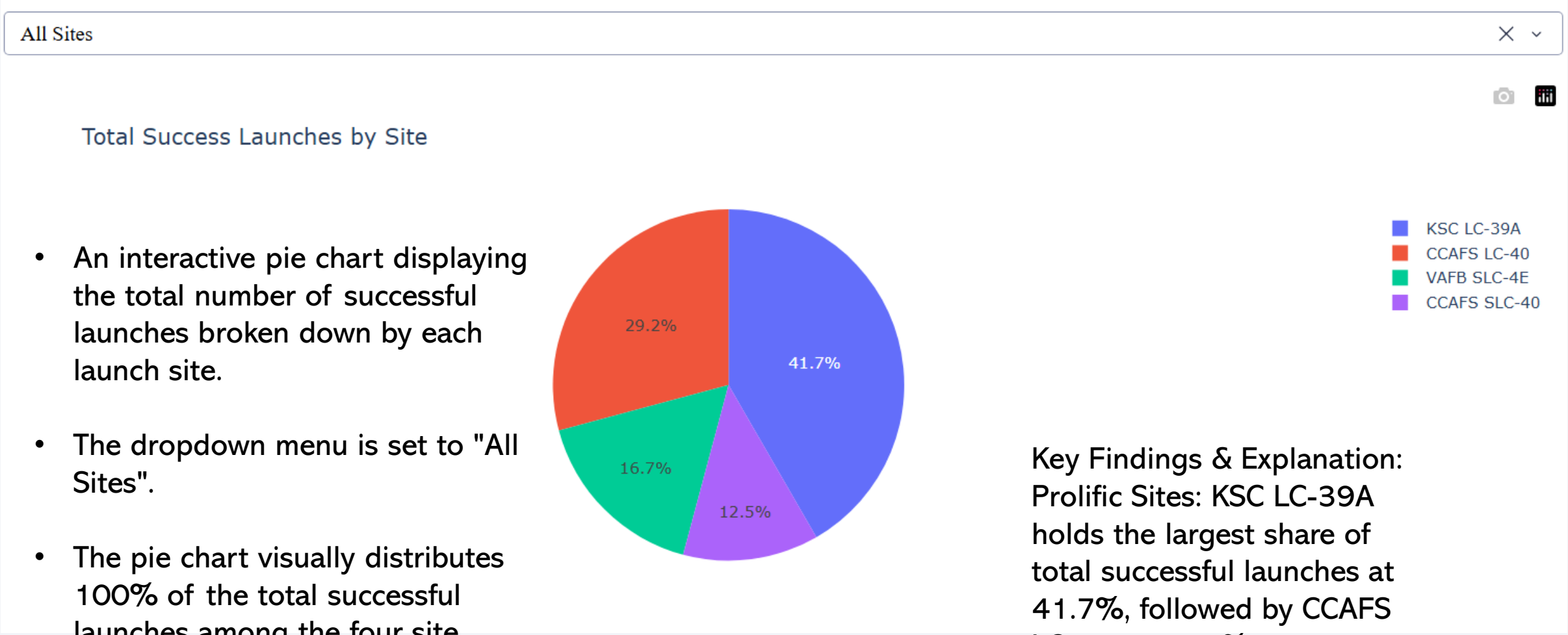
- Distance Markers: Displays the exact calculated distance (e.g., 0.86 KM to the coastline).



Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>



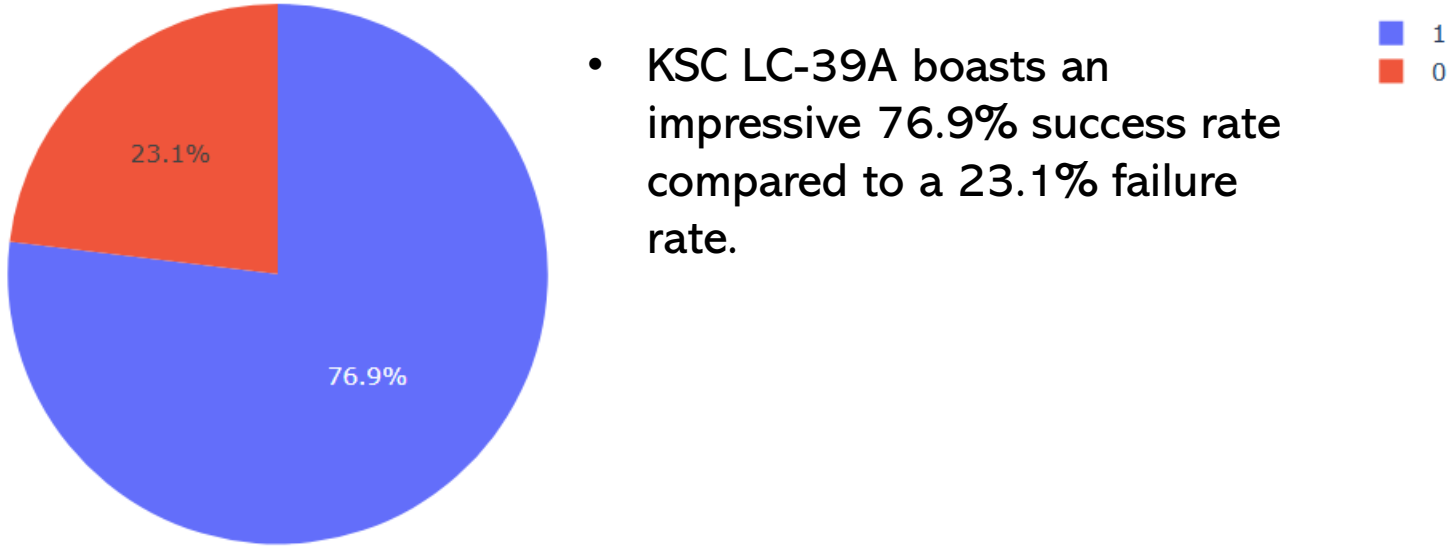
- An interactive pie chart displaying the total number of successful launches broken down by each launch site.
- The dropdown menu is set to "All Sites".
- The pie chart visually distributes 100% of the total successful launches among the four site designations.

Key Findings & Explanation:
Prolific Sites: KSC LC-39A holds the largest share of total successful launches at 41.7%, followed by CCAFS LC-40 at 29.2%.

<Dashboard Screenshot 2>

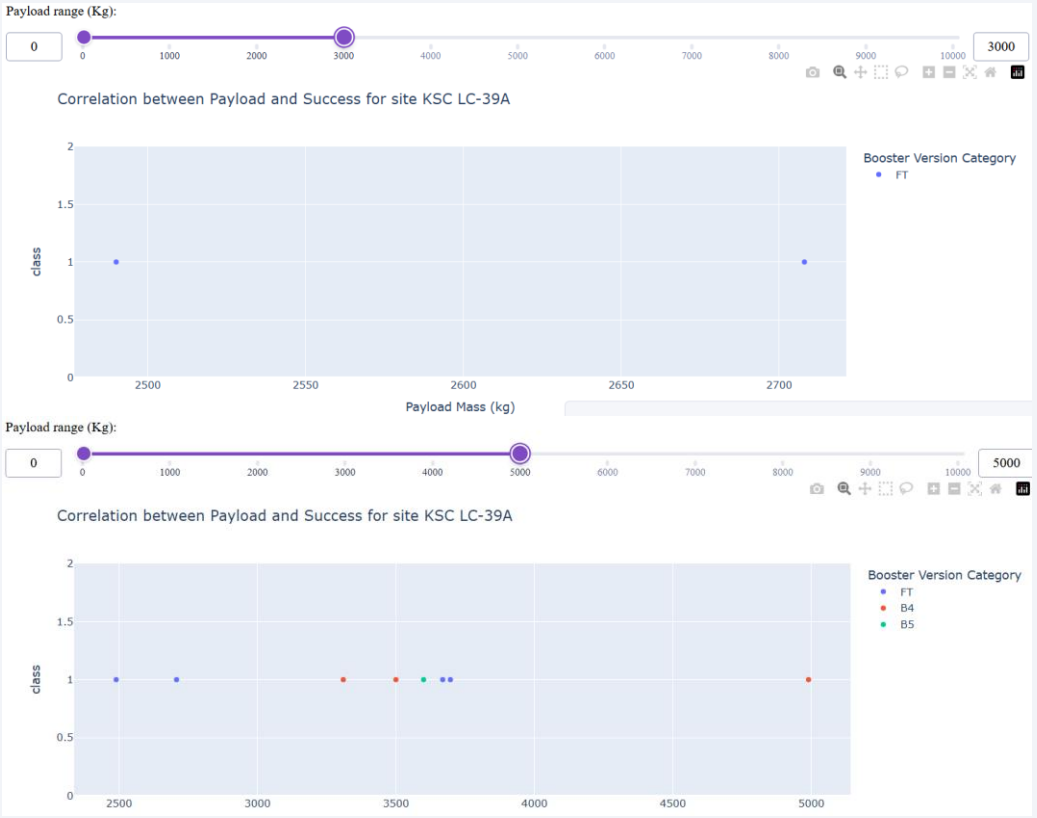
Total Success vs. Failure Launches for site KSC LC-39A

- A site-specific pie chart generated by selecting "KSC LC-39A" in the dropdown menu, detailing its specific ratio of successful to failed launches.
- The chart splits the site's outcomes into Class 1 (Success - blue) and Class 0 (Failure - red).



- KSC LC-39A boasts an impressive 76.9% success rate compared to a 23.1% failure rate.

<Dashboard Screenshot 3>

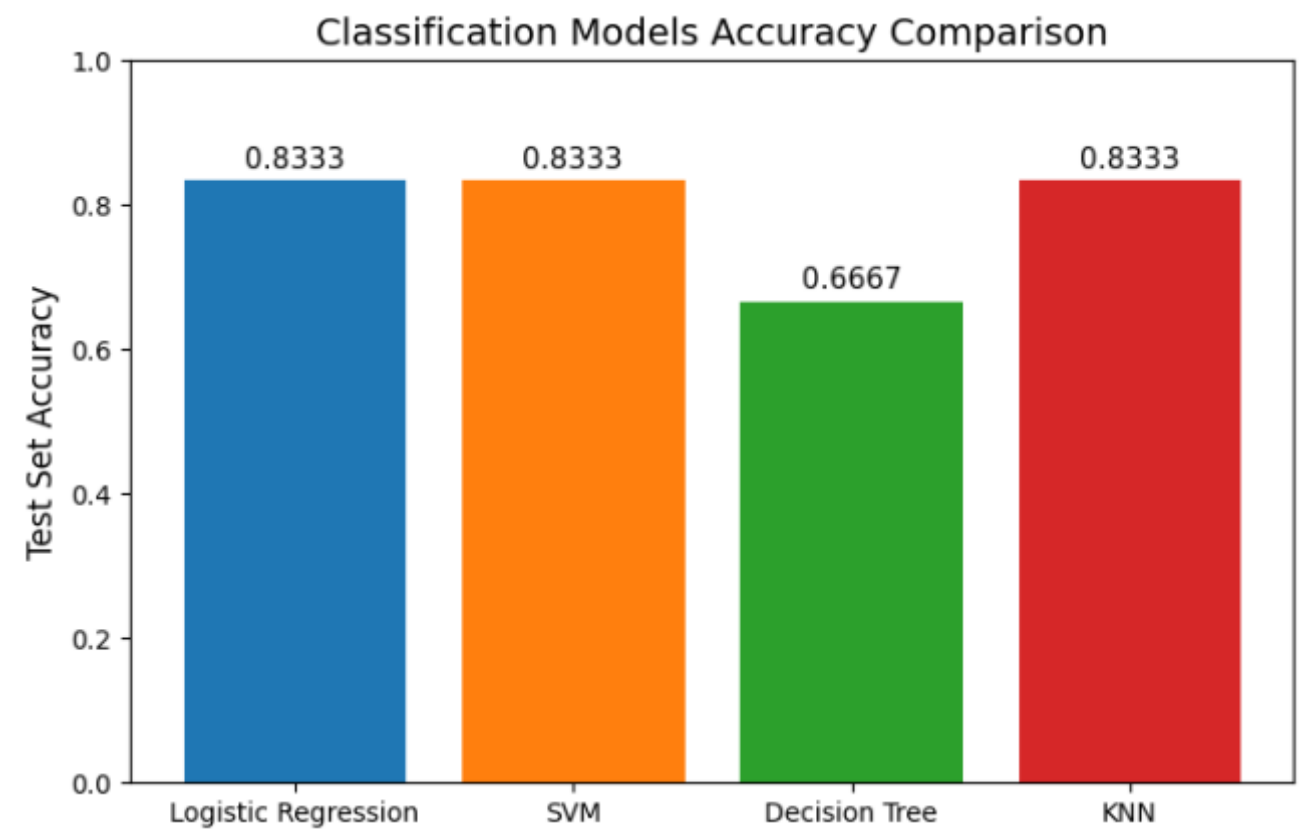


A series of scatter plots demonstrating how the interactive range slider dynamically filters the launch data based on payload mass (kg). Adjusted to display ranges of 0-7000 kg, 0-5000 kg, 0-4000 kg, and 0-3000 kg. The X-axis represents Payload Mass, the Y-axis is the outcome Class (0 for failure, 1 for success), and the data points are color-coded by the Booster Version Category (FT, B4, B5).

Section 5

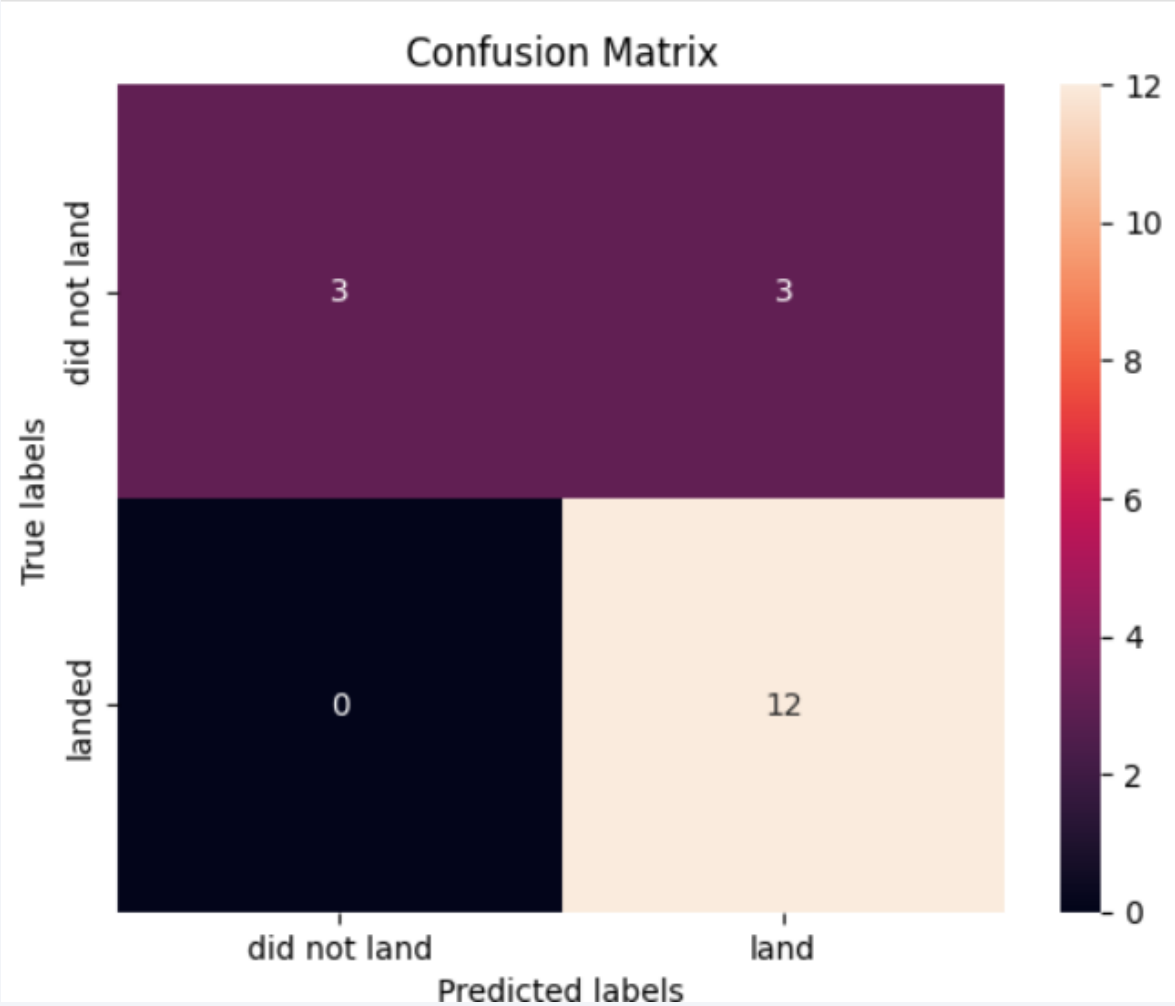
Predictive Analysis (Classification)

Classification Accuracy



- A bar chart comparing the test set accuracy of your four trained machine learning models.
- Three models (Logistic Regression, SVM, and KNN) share the exact same high score of 83.33%.
- The Decision Tree model shows a noticeably lower score of 66.67%.
- Any of the top three models would be highly capable of predicting Falcon 9 landing outcomes for cost estimation purposes.

Confusion Matrix



- Logistic Regression, SVM, and KNN model tests received a high 83.33% accuracy. Those 3 models had the same confusion matrix output.

- True Positives (Bottom Right - 12): The model correctly predicted a successful landing 12 times.
- True Negatives (Top Left - 3): The model correctly predicted a failed landing 3 times.
- False Positives (Top Right - 3): The model predicted the rocket would land, but it actually failed.
- False Negatives (Bottom Left - 0): The model predicted a failure, but it actually landed.

Conclusions

SpaceX has shown a dramatic, quantifiable improvement in success rates over the past decade, effectively mastering first-stage landing technology from 2015 onward.

- Payload mass, orbit type (e.g., LEO vs. GTO), and the chosen launch site are strong, measurable indicators of a Falcon 9 first stage's landing success.
- Geospatial Strategy: Launch sites are strategically placed near coastlines for flight safety (dropping stages over the ocean) and immediately adjacent to transport infrastructure (railways/highways) to optimize logistical efficiency.
- Machine learning models (specifically Logistic Regression, SVM, and KNN) are highly effective at predicting landing outcomes with 83.33% accuracy, providing a reliable quantitative tool for competitors to estimate SpaceX's launch costs.

Thank you!